

Response to Reviewers

Spatial Cluster Randomized Trials — Sampling Design with Spillover Effects & Spatial Dependence
BMC Medical Research Methodology — First Revision
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April 2026

Dear Editors and Reviewers,

We sincerely thank the Editor and both Reviewers for their thorough and constructive feedback. The comments have helped us substantially strengthen the manuscript. We have carefully addressed each point below, with all changes implemented in the revised manuscript file (Revisions_V1.tex). The original main.tex has been left unmodified for reference. Below we describe each change in detail, including the section location, line numbers in the revised file, and the opening text of each changed passage to facilitate cross-referencing with the annotated manuscript (Revisions_Annotated_V1.pdf, where all new or revised text is highlighted in red).

EDITOR COMMENTS

E1 — Declarations Section (Required)

Reviewer Comment	Author Response	Location & Evidence
The declarations section does not conform to BMC Medical Research Methodology formatting. The journal requires a formal Declarations section with seven subsections: Ethics Approval and Consent to Participate; Consent for Publication; Availability of Data and Materials; Competing Interests; Funding; Authors’ Contributions; Acknowledgements.	Restructured the entire back-matter under a single Declarations heading with all 7 required BMC subsections: Ethics Approval, Consent for Publication, Data Availability, Competing Interests, Funding, Authors’ Contributions, and Acknowledgements. The previous scattered subsection entries have been consolidated. Note: as part of post-review editorial revisions, the sentence “Summarized findings are included in Tables 1–3.” has also been removed from the Data Availability subsection, as it is redundant with the manuscript body.	Declarations section (end of manuscript). Lines: Revisions_V1.tex: ~1266–1292 Starts with: “\section*{Declarations}”

REVIEWER 1 COMMENTS

R1.1 — BSS Literature and McCann et al. (2018) (Major)

Reviewer Comment	Author Response	Location & Evidence
A key reference is missing: McCann et al. (2018) describes an algorithm for sampling and randomization of clusters to reduce spillover risk in CRTs. Differences and similarities with that work need to be discussed.	Added McCann et al. (2018) to the reference list and added a new paragraph contextualizing BSS relative to existing work. The paragraph cites McCann et al. (2018) as an example of spatially stratified allocation in field trials and explicitly states that the formal comparison of SRS vs. BSS under a SAR model with simultaneous spillover effects — and the exhaustive enumeration approach — is a novel contribution not present in prior literature. Note: as part of post-review editorial revisions, these two paragraphs were relocated from Section 2.4 (Methods) to Section 5.1 ‘Relevance of Results’ (Discussion), where they better contextualize the study’s contributions relative to prior work. They appear in red in the annotated manuscript in that location.	Discussion, Section 5.1 ‘Relevance of Results’ (relocated from Section 2.4 in Methods; see Post-Review Editorial Revisions below). Lines: Now in Discussion §5.1 (relocated) Starts with: “The spatial separation principle underlying BSS has antecedents in spatially balanced experimental designs.”

R1.2 — Distinguishing ρ (nuisance) from ψ (causal spillover) (Major)

Reviewer Comment	Author Response	Location & Evidence
The manuscript does not clearly distinguish between the spatial autocorrelation parameter ρ (a nuisance background parameter) and the spillover effect ψ (a causal design-level estimand). This distinction should be explained and collinearity limitations addressed.	Two additions were made: (1) A new paragraph was inserted before Section 2.6 explicitly distinguishing ρ as background nuisance spatial correlation from ψ as a causal spillover estimand, including consequences of failing to include both in the model. (2) Section 5.2 (Limitations) expands the collinearity note to clarify the identification challenge when both ρ and ψ are large. Note: as a post-review editorial revision, the sentence opening “Before describing the spatial weights matrix,” has been removed; the paragraph now begins directly with “It is important to clarify the conceptual distinction...”	New paragraph before Section 2.6 (Spatial Dependence); expanded collinearity note in Section 5.2 (Limitations). Lines: Revisions_V1.tex: ~538–553 Starts with: “It is important to clarify the conceptual distinction between two spatial quantities in this model...”

R1.3 — Inferential Scope Clarification (Major)

Reviewer Comment	Author Response	Location & Evidence
It is unclear how one can have valid frequentist inference for BSS models with only 2 possible allocations. The inferential scope of the simulation needs to be stated clearly.	Added a new paragraph before Section 3.2 clarifying that: (1) BSS is a design-stage tool evaluated through exhaustive enumeration, not a randomization distribution for permutation inference; (2) model-based inference via SAR MLE (specifically the lagsarlm() R function) is the appropriate framework for practitioners; (3) the simulation characterizes estimator performance across the full design space. An inference constraint note was also added to Section 5.2. Note: as post-review editorial revisions, the sentence “Before describing the simulation procedure, it is important to clarify the inferential scope of this study.” was	New paragraph before Section 3.2 (Simulation Procedure); additional note in Section 5.2 (Limitations). Lines: Revisions_V1.tex: ~669–681 Starts with: “BSS is fundamentally a design-stage tool: the set of valid BSS allocations is

	removed (paragraph now begins “BSS is fundamentally a design-stage tool...”), and “lagsarlm()” was updated to specify “lagsarlm() R function” for clarity.	determined prior to data collection...”
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R1.4 — McCann et al. (2018) Discussion (Minor)

Reviewer Comment	Author Response	Location & Evidence
McCann et al. (2018) should be cited and discussed in relation to the BSS approach.	Addressed as part of R1.1. McCann et al. (2018) is cited in the BSS novelty paragraph relocated to Discussion §5.1. The paragraph notes the connection to spatially stratified allocation and clarifies how the current work extends that framework into SAR modeling with explicit spillover.	See R1.1. Discussion, Section 5.1 ‘Relevance of Results’.

R1.5 — Risk-Management Framing for Results (Sections 3.4–3.6) (Major)

Reviewer Comment	Author Response	Location & Evidence
Sections 3.4, 3.5, and 3.6 claim BSS is generally superior in terms of MSE, which is not supported by the data. The framing needs to reflect a balanced risk-management perspective distinguishing average MSE from variance across allocations.	All three results sections were rewritten. The new framing distinguishes between (a) average MSE across allocations and (b) SD of MSE across allocations. Core message: SRS can achieve a lower average MSE in certain regimes (particularly TrtSpill), but BSS bounds the worst-case allocation risk through near-zero SD (only 2–6 valid allocations). Specific numerical examples from Tables 1–3 are cited in both directions. The 3×4 TrtSpill anomaly (BSS MSE > SRS MSE with $p=0.01$) is explained as the ‘checkerboard paradox’ and cross-referenced to the Discussion.	Sections 3.4, 3.5, and 3.6 (Intervention Effect Estimates: 2×4, 3×3, and 3×4 grids). Cross-reference to Section 5.1. Lines: Revisions_V1.tex: ~763–775 (2×4); ~986–996 (3×4) Starts with: “...reveal a risk-management trade-off between SRS and BSS rather than a uniform advantage for either method...”

R1.6 — Minor Text Edit (Minor)

Reviewer Comment	Author Response	Location & Evidence
Minor wording issue in the methods section.	Addressed via the ρ vs. ψ distinction added in R1.2. The new paragraph clarifies the conceptual separation of these two spatial quantities throughout Section 2.5–2.6.	See R1.2. New paragraph before Section 2.6.

R1.7 — Boundary-Level Spillover Limitation (Minor)

Reviewer Comment	Author Response	Location & Evidence
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The model applies spillover at the cluster level uniformly. In practice, subjects near cluster boundaries may experience stronger spillover than those near the center. This limitation should be acknowledged.	Added a new paragraph to Section 5.2 (Limitations) explicitly noting the cluster-level uniformity limitation: all subjects in a cluster receive the same binary spillover indicator regardless of their position within the cluster. Future work exploring boundary-level or subject-level spillover mechanisms is suggested.	New paragraph in Section 5.2 (Limitations). Lines: Revisions_V1.tex: ~1220–1222 Starts with: “An additional notable limitation is the rudimentary spillover mechanism (binary effect) applied...”
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R1.8 — Inference Constraint Note (Minor)

Reviewer Comment	Author Response	Location & Evidence
The paper should address the inference constraint that arises from the restricted BSS randomization space.	Addressed as part of R1.3. A brief note on the inference constraint under BSS (very limited randomization space precludes permutation-based testing) was added to Section 5.2 Limitations.	See R1.3. Section 5.2 (Limitations). Lines: Revisions_V1.tex: ~1224–1228 Starts with: “Because this study evaluates estimator properties through exhaustive enumeration of all possible treatment allocations...”

R1.9 — BSS Variance Across Allocations (Minor)

Reviewer Comment	Author Response	Location & Evidence
The discussion of BSS performance should acknowledge that the very small number of BSS allocations makes variance estimation and interpretation different from SRS.	Addressed as part of the risk-management rewrite in R1.5. Sections 3.4–3.6 now explicitly discuss BSS SD across allocations as a key design consideration, noting that the 2–6 valid BSS allocations produce near-zero allocation variance, which bounds worst-case performance.	See R1.5. Sections 3.4, 3.5, 3.6.

REVIEWER 2 COMMENTS

R2.1 — Move Technical Content to Supplementary Appendix (Major)

Reviewer Comment	Author Response	Location & Evidence
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The manuscript is dense with mathematical derivations and procedural detail that impede readability. Consider moving some technical content to a supplementary appendix.	Five blocks of content were moved to new appendices (Appendices A–E in Revisions_V1.tex), with bridging notes in the main text: (A) Spatial weights matrix equations from §2.6; (B) Full SAR log-likelihood derivation from §2.7; (C) Complete 8-step simulation algorithm from §3.2; (D) Moran’s I and Geary’s C formulas from §2.6; (E) Distance-based neighbor construction from §2.2.2. Each bridging note retains a conceptual description and directs the reader to the appendix.	Appendices A–E (end of manuscript, after Bibliography). Bridging notes in Sections 2.2, 2.6, 2.7, 3.2. Lines: Revisions_V1.tex: ~1298–1453 (Appendices A–E, entirely new) Starts with: “Lines 1298–1453 are entirely new in Revisions_V1.tex (did not exist in main.tex)”
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R2.2 — Additional References for BSS Terminology (Major)

Reviewer Comment	Author Response	Location & Evidence
The term ‘block stratified sampling’ is unfamiliar in the context used. More references are needed and the connection to standard geographic stratification in CRTs should be made explicit.	Two additions were made. First, a sentence with geographic stratification citations was added at the BSS definition in Section 2.4.1, connecting BSS to the broader literature on geographic stratification in CRTs (McCann et al. 2018; Crisp et al. 2023). The sentence explicitly notes that the spatial isolation condition (no shared Rook edges) is a novel constraint. Second, a novelty paragraph further elaborates on how this work extends prior spatially balanced allocation approaches. Note: these novelty paragraphs were subsequently relocated to Discussion §5.1 (see R1.1 and Post-Review Editorial Revisions).	Section 2.4.1 BSS description (citations added). Novelty paragraphs relocated to Discussion §5.1. Lines: Revisions_V1.tex: ~356–372 (BSS description) Starts with: “Block stratified sampling ensures a balanced number of intervention and control cells are selected, with the added condition...”

R2.3 — Clarify BSS Novelty vs. Established Methods (Major)

Reviewer Comment	Author Response	Location & Evidence
Is the no-common-edge-boundary condition novel or standard? References and explicit clarification of what is adapted from prior work are needed.	The BSS novelty paragraph explicitly states that geographic stratification is established in the CRT literature, but that the comparison of SRS vs. BSS under a SAR model with explicit spillover effects — and the exhaustive enumeration of the full design space — is a novel contribution. The spatial isolation condition (no shared edges between same-treatment clusters) is identified as novel. McCann et al. (2018) and Crisp et al. (2023) are cited as related but methodologically distinct work.	See R1.1/R2.2. Discussion, Section 5.1 (relocated).

R2.4 — Best-Effort BSS in Real-World NC Application (Major)

Reviewer Comment	Author Response	Location & Evidence
Figure 10 is described as BSS but the no-common-edge condition is not fully satisfied. The text should clarify that real-world geographies require a best-effort approach.	Added explicit language at the end of Section 4.4 clarifying that Figure 10 represents a ‘best-effort’ block stratification for the NC judicial district geography, where irregular shapes and varying numbers of shared borders mean the strict no-common-edge criterion cannot always be satisfied. Investigators are advised to minimize shared borders and document deviations from the ideal. Note: as a post-review editorial revision, ‘Block Stratification sampling’ was corrected to ‘block stratified sampling’ in this paragraph.	Section 4.4 (Recommendations for Treatment Assignment), end of application paragraph. Lines: Revisions_V1.tex: ~1189–1201 Starts with: “...represents a ‘best-effort’ block stratification: the assignment minimizes shared borders between same-treatment districts...”

R2.5 — MSE = Bias² + Variance Decomposition (Major)

Reviewer Comment	Author Response	Location & Evidence
Figure 5 has an unusual appearance. Which patterns reflect bias versus variance? The MSE decomposition should be made explicit.	Added a sentence to Section 3.3 (Simulation Evaluation Metrics) stating that $MSE = Bias^2 + Variance$, so differences in MSE between SRS and BSS can be attributed to either systematic bias or sampling variability across allocations, and that this decomposition is visually apparent in the grouped bar figures throughout the Results section.	Section 3.3 (Simulation Evaluation Metrics). Lines: Revisions_V1.tex: ~756–757 Starts with: “Since $MSE = Bias^2 + Variance$, differences in MSE between SRS and BSS can be attributed to either systematic bias or sampling variability...”

R2.6 — Covariate-Constrained Randomization Literature (Major)

Reviewer Comment	Author Response	Location & Evidence
The paper is missing key background literature on covariate-constrained randomization. The absence of any description of covariate-constrained randomization is notable. Crisp et al. (2023) is one example.	Added a paragraph discussing covariate-constrained randomization. The paragraph cites Crisp et al. (2023) and explicitly distinguishes BSS from covariate-constrained approaches: BSS restricts allocations based on spatial adjacency (no shared Rook edges), whereas covariate-constrained randomization restricts based on measured covariate balance. The two approaches are noted as complementary, with covariate-balance extensions suggested as future work. Note: as a post-review editorial revision, this paragraph was relocated to Discussion §5.1 (see R1.1 and Post-Review Editorial Revisions).	Discussion, Section 5.1 ‘Relevance of Results’ (relocated from Methods §2.4). Lines: Now in Discussion §5.1 (relocated) Starts with: “A related line of work concerns covariate-constrained randomization, in which the set of eligible allocations is restricted to

		those achieving covariate balance..."
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R2.7 — ‘Prevention’ → ‘Limitation’ of Treatment Contamination (Minor)

Reviewer Comment	Author Response	Location & Evidence
‘Prevention of treatment contamination’ is too strong — CRTs limit contamination, they do not prevent it entirely.	Changed ‘prevention of treatment contamination’ to ‘limitation of treatment contamination’ in Section 2.1.	Section 2.1 (Cluster Randomized Trials), first substantive paragraph.

R2.8 — Direct vs. Total Effect Clarification (Minor)

Reviewer Comment	Author Response	Location & Evidence
What the authors call a ‘direct effect’ is technically a ‘total effect’ per Halloran et al., as it captures both direct and indirect effects for treated individuals in CRTs.	Added a clarifying parenthetical at the end of the intervention effect definition in Section 2.3.1: when spillover propagates back into intervention clusters (TrtSpill), the estimated β captures elements of both direct and indirect effects and is more precisely interpreted as a total treatment effect.	Section 2.3.1 (Intervention Effect definition). Lines: Revisions_V1.tex: ~283 Starts with: "...in such scenarios β may capture elements of both the direct intervention effect and the indirect spillover..."

R2.9 — Figure 2b Arrow Direction and Caption (Minor)

Reviewer Comment	Author Response	Location & Evidence
Figure 2b is not clear. Why are there no arrows from the upper-right red block? The caption should be lengthened to explain spillover directionality.	Two corrections were made. First, the Figure 2 image was corrected: the top row panel (b) had intervention (red) and control (blue) cells swapped. The colors have been fixed so that panel (b) correctly depicts the TrtSpill pattern. Second, the caption was updated to explicitly describe spillover directionality: panel (a) TrtNoSpill shows arrows flowing only from intervention cells into neighboring control cells; panel (b) TrtSpill shows arrows flowing into both control and intervention neighbors.	Figure 2 image (TrtSpillComparisonV2.png) corrected; Figure 2 caption updated. Lines: Revisions_V1.tex: ~347–349 Starts with: "(a) TrtNoSpill: spillover arrows flow outward from intervention cells (red) into neighboring control cells (blue) only..."

R2.10 — Checkerboard Paradox in Discussion (Minor)

Reviewer Comment	Author Response	Location & Evidence
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Doesn't the checkerboard pattern maximize potential spillover by surrounding all intervention clusters on all sides? This apparent paradox should be acknowledged.	Added a paragraph to Section 5.1 naming and explaining the 'checkerboard paradox': BSS's spatial isolation feature becomes a liability under TrtSpill + $\rho > 0$ because every intervention cluster is surrounded exclusively by control neighbors, amplifying spillover-induced bias. Recommendation: consider prior knowledge of the spillover mechanism when choosing between SRS and BSS.	Section 5.1 (Relevance of Results), new paragraph. Lines: Revisions_V1.tex: ~1212–1218 Starts with: "An important nuance observed in the 3×4 simulation results warrants attention in the context of practical design recommendations..."
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R2.11 — Y_{ik} Notation Clarification (Minor)

Reviewer Comment	Author Response	Location & Evidence
Y_{ik} notation uses dual subscripts (i = subject, k = cluster) but the description 'k and k prime are different cluster indices' is confusing.	Revised notation description to: 'where i indexes individual subjects, k denotes the cluster to which subject i belongs, k' indexes a neighboring cluster, ρ is the spatial correlation coefficient, and ϵ_i is an i.i.d. error term.'	Section 2.5 (SAR Model), below Equation 1.

R2.12 — Z_i Indicator Explanation (Minor)

Reviewer Comment	Author Response	Location & Evidence
How is z_i determined? What does 'can experience any spillover effect' mean in plain language?	Updated the definition to: ' z_i is an indicator of whether subject i belongs to a cluster that is a Rook neighbor of at least one intervention cluster, making that subject eligible to receive spillover.'	Section 2.5 (SAR Model), paragraph defining all model terms.

POST-REVIEW EDITORIAL REVISIONS

In addition to the 22 changes described above, the following six editorial improvements were made to the revised manuscript. These changes address clarity, terminology, and document flow issues identified during the revision process. Each is noted in the annotated manuscript (Revisions_Annotated_V1.pdf) where applicable.

Edit	Description	Location in Manuscript
Edit 1	Relocated the BSS novelty paragraph (McCann et al. 2018) and the covariate-constrained randomization paragraph from the end of Methods §2.4.3 to Discussion §5.1 'Relevance of Results', where they more naturally contextualize the study's contributions. Both paragraphs remain	Removed from: Methods §2.4.3 (3×4 Grid). Added to: Discussion §5.1 (after opening paragraph, before 'Our findings show...').

	highlighted in red in the annotated PDF at their new location.	
Edit 2	Removed the forward-referencing opener ‘Before describing the spatial weights matrix,’ from the start of the ρ vs. ψ distinction paragraph (R1.2). Paragraph now begins directly with ‘It is important to clarify the conceptual distinction...’	Before Section 2.6 (Spatial Dependence). R1.2 paragraph.
Edit 3	Removed the forward-referencing opening sentence ‘Before describing the simulation procedure, it is important to clarify the inferential scope of this study.’ from the R1.3 paragraph. The paragraph now opens directly with ‘BSS is fundamentally a design-stage tool...’	Before Section 3.2 (Simulation Procedure). R1.3 paragraph.
Edit 4	Updated ‘specifically the lagsarlm() function’ to ‘specifically the lagsarlm() R function’ to make clear this is an R package function from the spdep library.	R1.3 paragraph (before Section 3.2).
Edit 5	Corrected ‘Block Stratification sampling’ to ‘block stratified sampling’ for consistent terminology throughout the manuscript.	Section 4.4 (Recommendations for Treatment Assignment / NC Application paragraph).
Edit 6	Removed the sentence ‘Summarized findings are included in Tables 1–3.’ from the Availability of Data and Materials subsection of Declarations. This sentence is redundant with content in the manuscript body.	Declarations > Availability of Data and Materials subsection.

SUMMARY OF ALL CHANGES

Item	Source	Type	Description	Status / Location in Revised MS
E1	Editor	Required	Restructure Declarations section (7 BMC subsections)	Done — Declarations section
R1.1	Reviewer 1	Major	BSS literature grounding; add McCann et al. (2018)	Done — Discussion §5.1 (relocated from §2.4)
R1.2	Reviewer 1	Major	ρ vs. ψ distinction; collinearity limitation	Done — Before §2.6; §5.2
R1.3	Reviewer 1	Major	Inferential scope clarification for BSS	Done — Before §3.2; §5.2
R1.4	Reviewer 1	Minor	McCann et al. (2018) discussion	Done — See R1.1

R1.5	Reviewer 1	Major	Rewrite §3.4–3.6 with risk-management framing	Done — Sections 3.4, 3.5, 3.6
R1.6	Reviewer 1	Minor	Minor text edit (addressed via R1.2)	Done — See R1.2
R1.7	Reviewer 1	Minor	Boundary-level spillover limitation paragraph	Done — Section 5.2
R1.8	Reviewer 1	Minor	Inference constraint note	Done — Section 5.2 (see R1.3)
R1.9	Reviewer 1	Minor	BSS allocation variance framing (via R1.5)	Done — Sections 3.4–3.6
R2.1	Reviewer 2	Major	Move 5 technical blocks to Appendices A–E	Done — Appendices A–E added
R2.2	Reviewer 2	Major	Geographic stratification citations + novelty paragraph	Done — §2.4.1; novelty paras in Discussion §5.1
R2.3	Reviewer 2	Major	Clarify BSS novelty vs. established methods	Done — See R1.1/R2.2
R2.4	Reviewer 2	Major	Best-effort BSS note for NC application	Done — Section 4.4
R2.5	Reviewer 2	Major	MSE = Bias ² + Variance decomposition sentence	Done — Section 3.3
R2.6	Reviewer 2	Major	Covariate-constrained randomization; cite Crisp et al. (2023)	Done — Discussion §5.1 (relocated from §2.4)
R2.7	Reviewer 2	Minor	prevention → limitation of treatment contamination	Done — Section 2.1
R2.8	Reviewer 2	Minor	Direct vs. total effect clarification	Done — Section 2.3.1
R2.9	Reviewer 2	Minor	Figure 2b colors corrected + caption updated	Done — Figure 2 image + caption
R2.10	Reviewer 2	Minor	Checkerboard paradox note in Discussion	Done — Section 5.1
R2.11	Reviewer 2	Minor	Y_{ik} notation clarification	Done — Section 2.5
R2.12	Reviewer 2	Minor	Z_i indicator explanation	Done — Section 2.5
Edit 1	Post-review	Editorial	Relocated McCann/novelty paragraphs from Methods §2.4.3 to Discussion §5.1	Done — Discussion §5.1 (red in annotated MS)
Edit 2	Post-review	Editorial	Removed ‘Before describing the spatial weights matrix,’ opener from R1.2 paragraph	Done — Before §2.6
Edit 3	Post-review	Editorial	Removed forward-referencing opening sentence from R1.3 paragraph	Done — Before §3.2
Edit 4	Post-review	Editorial	Updated lagsarlm() to lagsarlm() R function for clarity	Done — Before §3.2

Edit 5	Post-review	Editorial	Fixed ‘Block Stratification sampling’ → ‘block stratified sampling’	Done — Section 4.4
Edit 6	Post-review	Editorial	Removed redundant ‘Summarized findings...’ sentence from Declarations	Done — Declarations > Data Availability

We hope these revisions address all concerns raised by the Editors and Reviewers, and we look forward to hearing the outcome of the review.

Respectfully submitted,

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